

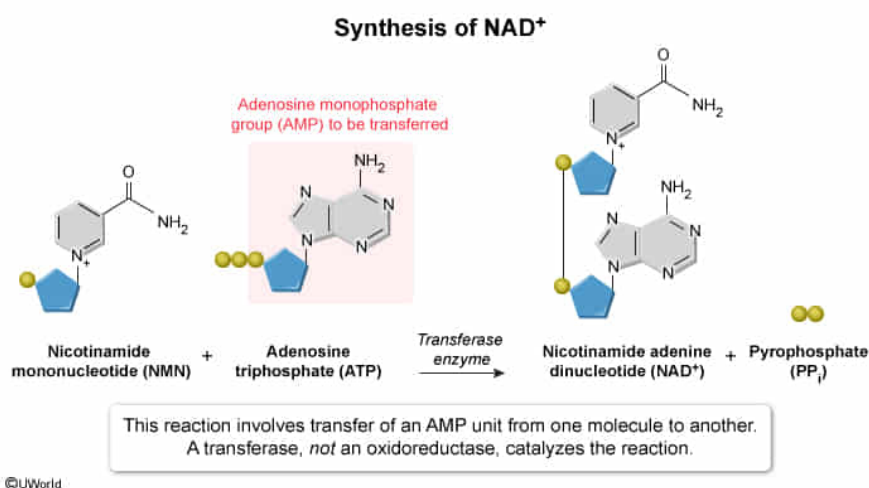
Which of the following enzyme-catalyzed reactions does NOT involve an oxidoreductase?

- ☐ A. Glucose-6-phosphate + NADP⁺ → 6-phosphogluconolactone + NADPH + H⁺ [5%]
- ☐ B. Cystine + NADH + H⁺ → 2 cysteine + NAD⁺ [5%]
- ☐ C. 4 Cytochrome C_{red} + O₂ + 4 H⁺ → 4 Cytochrome C_{ox} + 2 H₂O [14%]
- ✓ ☒ D. Nicotinamide mononucleotide + ATP → NAD⁺ + pyrophosphate [74%]

Correct

73% answered correctly

Explanation:



Enzymes are **classified** by the general types of reactions they catalyze. **Oxidoreductases** are enzymes that catalyze oxidation-reduction (redox) reactions. **Biological redox reactions often include the interconversion** of an

electron carrier between its **oxidized and reduced forms**. Common electron carriers include the oxidized and reduced forms of **cytochromes** (eg, cytochrome C_{ox} and cytochrome C_{red}) as well as the oxidized and reduced forms of several cofactors: **NAD⁺ and NADH, NADP⁺ and NADPH, or FAD and FADH₂**.

The question asks which of the reactions is NOT catalyzed by an oxidoreductase. The reaction shown in Choice D depicts the synthesis of nicotinamide adenine dinucleotide (NAD⁺) from nicotinamide mononucleotide (NMN) and adenosine triphosphate (ATP). Although this reaction involves NAD⁺, it does *not* involve conversion of NAD⁺ to NADH or vice versa; thus, it is *not* a redox reaction.

Instead, it is a *transfer* reaction in which an adenosine monophosphate (AMP) group is transferred from ATP to NMN, leaving pyrophosphate (PP_i). Therefore, this reaction is catalyzed by a **transferase**, *not* an oxidoreductase.

(Choice A) This reaction depicts the first step in the oxidative phase of the **pentose phosphate pathway**, in which glucose-6-phosphate becomes oxidized and NADP⁺ is reduced to NADPH. Because it is a redox reaction, the enzyme involved must be an oxidoreductase.

(Choice B) This reaction shows the reduction of **cystine** (ie, two cysteine molecules linked by a disulfide bond),

with NADH oxidized to NAD^+ in the process. Because it is a redox reaction, an oxidoreductase enzyme is required.

(Choice C) This reaction depicts the final step of the [electron transport chain](#), in which the reduced form of cytochrome C (cytochrome C_{red}) becomes oxidized and oxygen gas is reduced to water in the process. The reaction requires an oxidoreductase enzyme.

Educational objective:

Oxidoreductases are enzymes that catalyze oxidation-reduction (redox) reactions. They commonly interconvert the oxidized and reduced forms of cytochrome proteins or the cofactors NAD^+ and NADH, NADP^+ and NADPH, or FAD and FADH_2 .

Time spent:0 sec

Subject:
Biochemistry

QID:403243

System:
Enzymes

The cleavage of fructose 1,6-bisphosphate into glyceraldehyde-3-phosphate and dihydroxyacetone phosphate is thermodynamically unfavorable, with a $\Delta G'^{\circ}$ of +23.8 kJ/mol. However, in most cells the reaction readily proceeds to the right. Which of the following best explains this observation?

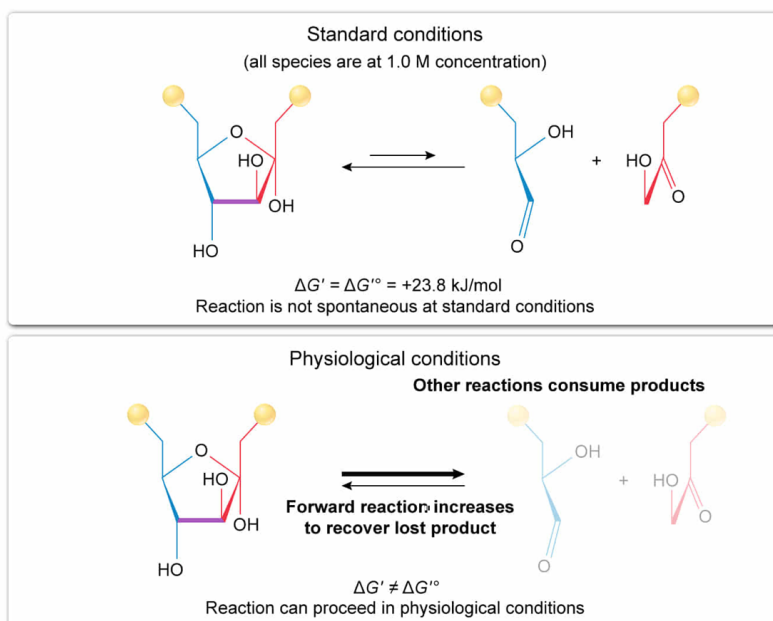
- ☐ A. This reaction is coupled to the hydrolysis of ATP *in vivo*. [36%]
- ☐ B. This is a multistep reaction with an exergonic elementary step. [13%]
- ✓ ☒ C. The products quickly participate in other reactions. [24%]
- ☐ D. The enzyme that catalyzes this reaction lowers the activation energy to shift equilibrium. [25%]

Correct

24% answered correctly

Explanation:

$\Delta G'^{\circ}$ at standard conditions may not reflect $\Delta G'$ at physiological conditions



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The change in Gibbs free energy $\Delta G'$ of a reaction can be taken as a measure of the reaction's **spontaneity**. The value of $\Delta G'$ can be calculated from the equation:

$$\Delta G' = \Delta G'^{\circ} + RT \ln Q$$

where $\Delta G'^{\circ}$ is the value of $\Delta G'$ at **standard conditions** (ie, 1 M all reactants and products, pH = 7), R is the **ideal gas constant**, T is the **temperature in kelvin**, and Q is the **reaction quotient**. The equation shows that $\Delta G'$ **varies with the concentrations** of the reactants and products involved. However, metabolite concentrations can vary with tissue type or physiological state, so different cells may have different $\Delta G'$ values for the same reaction.

Because of this variance, the standard free energy change $\Delta G'^{\circ}$ is often used as a convenient point of reference.

Nevertheless, *physiological* conditions in the cell may cause $\Delta G'$ to vary significantly from $\Delta G'^{\circ}$. In this question, **consumption of the products** by other reactions decreases the value of Q , causing the second term of the equation ($RT \ln Q$) to be **sufficiently negative** to overcome the positive value of $\Delta G'^{\circ}$. This shifting of equilibrium upon removal of the products is an application of **Le Châtelier's principle**.

(Choice A) The cleavage of fructose-1,6-bisphosphate is achieved by the enzyme aldolase during glycolysis. The energy-investment phase of glycolysis uses ATP with only two enzymes (hexokinase and phosphofructokinase-1); aldolase does not use ATP.

(Choice B) Gibbs free energy is a **state function** whose value is independent of the path taken (ie, single-step or multistep). If the overall free energy change is positive, the reaction will be

nonspontaneous whether proceeding in one step or multiple steps.

(Choice D) Although enzymes and catalysts lower the **energy of activation**, they do *not* alter equilibrium nor do they alter the free energy of products or reactants.

Educational objective:

The physiological value of the change in Gibbs free energy $\Delta G'$ can vary significantly from the reference value of $\Delta G'^{\circ}$.

Changing concentrations of products or reactants can shift reactions in the forward or reverse direction and the $\Delta G'$ value will change to reflect the new equilibrium.

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Biochemistry

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